Diploma Thesis

Investigation of Machine Vibration Analysis System Design

Vasileios Dimitriou

Department of Electrical and Software Engineering
University of Thessaly

September, 2025

1 Introduction

2 Background

3 Engineering and Design

4 Implementation

5 Results

6 Conclusions

1 Introduction
Predictive Maintenance and Condition Monitoring
Problem Requirements
Objective of the Thesis

2 Background

3 Engineering and Design

4 Implementation

5 Results

6 Conclusions

1 Introduction
Predictive Maintenance and Condition Monitoring
Problem Requirements
Objective of the Thesis

2 Background

3 Engineering and Design

4 Implementation

5 Results

6 Conclusions

Predictive Maintenance and Condition Monitoring

- Aims at tracking the health status of machines.

Predictive Maintenance and Condition Monitoring

- Aims at tracking the health status of machines.
- Tracks the deterioration of temperature and vibration.

Predictive Maintenance and Condition Monitoring

- Aims at tracking the health status of machines.
- Tracks the deterioration of temperature and vibration.
- Alert maintenance to take action before it is to late.

1 Introduction
Predictive Maintenance and Condition Monitoring
Problem Requirements
Objective of the Thesis

2 Background

3 Engineering and Design

4 Implementation

5 Results

6 Conclusions

Problem Requirements

- Sensing entity for data collection.

Problem Requirements

- Sensing entity for data collection.
- Processing unit for data analysis.

Problem Requirements

- Sensing entity for data collection.
- Processing unit for data analysis.
- Space for data storage.

Problem Requirements

- Sensing entity for data collection.
- Processing unit for data analysis.
- Space for data storage.
- Cloud-based for remote access.

1. Introduction
   - Predictive Maintenance and Condition Monitoring
   - Problem Requirements
   - Objective of the Thesis
2. Background
3. Engineering and Design
4. Implementation
5. Results
6. Conclusions

Objective of the Thesis

- Development of a low-cost embedded system for predictive maintenance [see Figure 1].

Objective of the Thesis

- Development of a low-cost embedded system for predictive maintenance [see Figure 1].
- Development of software for data acquisition and transmission from Arduino UNO R3 to Raspberry Pi 4.

Objective of the Thesis

- Development of a low-cost embedded system for predictive maintenance [see Figure 1].
- Development of software for data acquisition and transmission from Arduino UNO R3 to Raspberry Pi 4.
- Development of software for FFT-based spectral analysis software on Raspberry Pi 4 for frequency-domain feature extraction.

Objective of the Thesis

- Development of a low-cost embedded system for predictive maintenance [see Figure 1].
- Development of software for data acquisition and transmission from Arduino UNO R3 to Raspberry Pi 4.
- Development of software for FFT-based spectral analysis software on Raspberry Pi 4 for frequency-domain feature extraction.
- Development of a secure, authenticated API client for structured data transmission from Raspberry Pi 4 to Supabase.

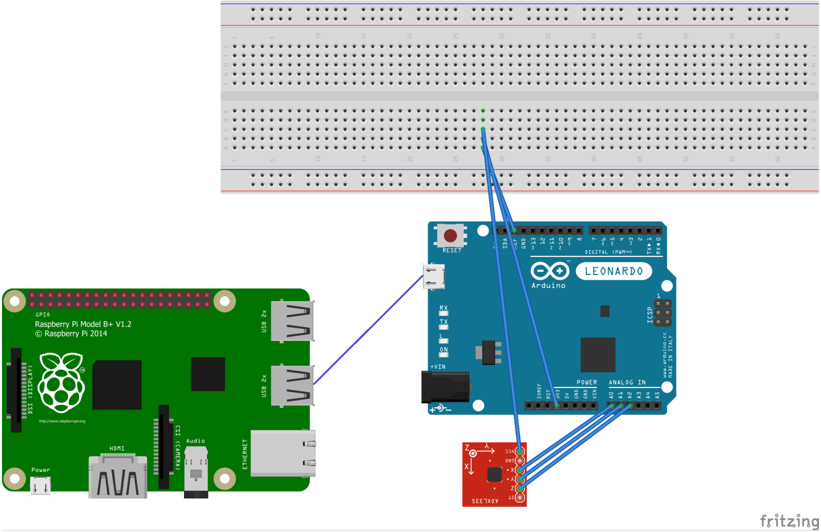
Objective of the Thesis

- Development of a low-cost embedded system for predictive maintenance [see Figure 1].
- Development of software for data acquisition and transmission from Arduino UNO R3 to Raspberry Pi 4.
- Development of software for FFT-based spectral analysis software on Raspberry Pi 4 for frequency-domain feature extraction.
- Development of a secure, authenticated API client for structured data transmission from Raspberry Pi 4 to Supabase.
- Implementation of a Vercel API endpoint that fetches measurement data from Supabase database.

Objective of the Thesis

- Development of a low-cost embedded system for predictive maintenance [see Figure 1].
- Development of software for data acquisition and transmission from Arduino UNO R3 to Raspberry Pi 4.
- Development of software for FFT-based spectral analysis software on Raspberry Pi 4 for frequency-domain feature extraction.
- Development of a secure, authenticated API client for structured data transmission from Raspberry Pi 4 to Supabase.
- Implementation of a Vercel API endpoint that fetches measurement data from Supabase database.
- Development of a React frontend that requests and visualizes data from the Vercel API.

Predictive Maintenance Tools



1 Introduction

2 Background
   Fast-Fourier Transform
   Fault Detection

3 Engineering and Design

4 Implementation

5 Results

6 Conclusions

1 Introduction

2 Background
Fast-Fourier Transform
Fault Detection

3 Engineering and Design

4 Implementation

5 Results

6 Conclusions

FFT

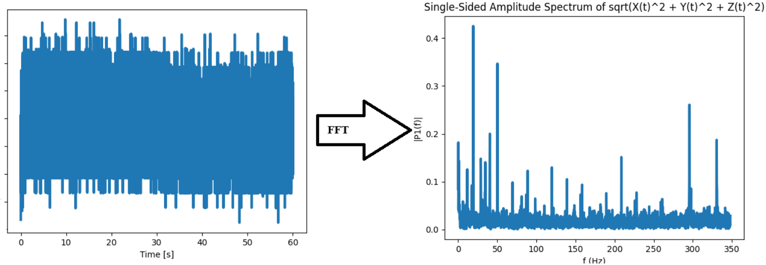


Figure 2: Conversion of the time-domain signal into its frequency-domain power representation using the FFT.

1. Introduction
2. Background
   - Fast-Fourier Transform
   - Fault Detection
3. Engineering and Design
4. Implementation
5. Results
6. Conclusions

Machinery Fault Tree Analysis

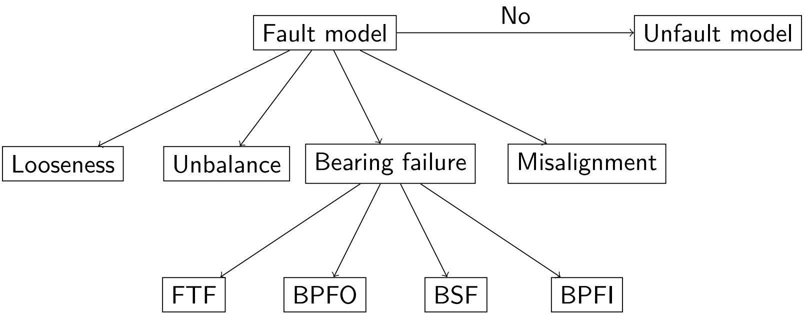


Figure 3: Fault tree analysis diagram showing the relationship between machine conditions and specific fault frequencies.

1 Introduction

2 Background

3 Engineering and Design

4 Implementation

5 Results

6 Conclusions

System Architecture and Dataflow

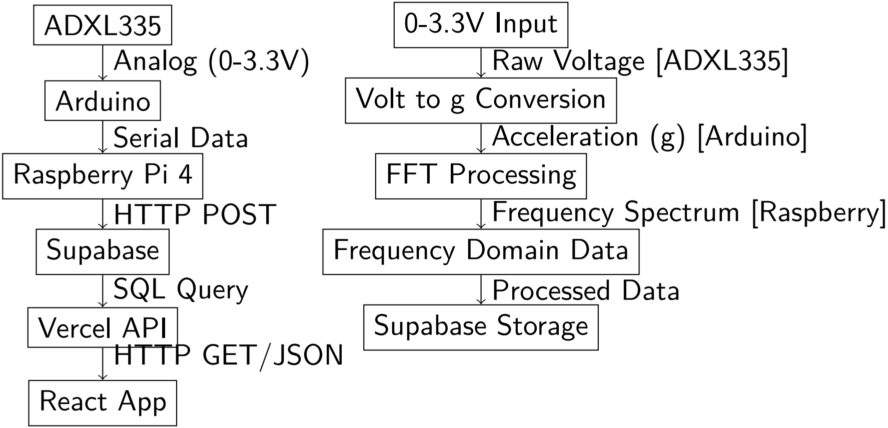


Figure 4: Complete system architecture and data flow and transformation pipeline from physical sensor to web visualization dashboard.

System boundaries

- The ADXL335 measurement range is $\pm 3\,g$.

System boundaries

- The ADXL335 measurement range is $\pm 3\,g$.
- The Arduino UNO R3 sampling rate is 700 [Hz].

System boundaries

- The ADXL335 measurement range is $\pm 3\,g$.
- The Arduino UNO R3 sampling rate is 700 [Hz].
- Flash memory and SRAM limitations require data to be transmitted in batches.

System boundaries

- The ADXL335 measurement range is $\pm 3\,g$.
- The Arduino UNO R3 sampling rate is 700 [Hz].
- Flash memory and SRAM limitations require data to be transmitted in batches.
- The frequency-domain analysis is limited to 350 Hz (Nyquist limit).

System boundaries

- The ADXL335 measurement range is $\pm 3\,g$.
- The Arduino UNO R3 sampling rate is 700 [Hz].
- Flash memory and SRAM limitations require data to be transmitted in batches.
- The frequency-domain analysis is limited to 350 Hz (Nyquist limit).
- The measurement time window determines the total number of samples collected.

1 Introduction

2 Background

3 Engineering and Design

4 Implementation
- Arduino UNO Algorithm
- Raspberry Pi Algorithm
- Vercel Implementation and Supabase Integration

5 Results

6 Conclusions

1 Introduction

2 Background

3 Engineering and Design

4 Implementation
- Arduino UNO Algorithm
- Raspberry Pi Algorithm
- Vercel Implementation and Supabase Integration

5 Results

6 Conclusions

Arduino UNO Algorithm

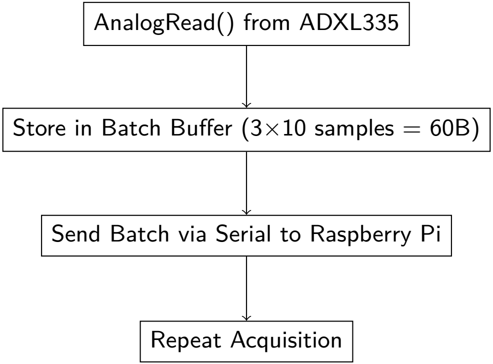


Figure 5: The Arduino UNO Algorithm.

1. Introduction
2. Background
3. Engineering and Design
4. Implementation
   - Arduino UNO Algorithm
   - Raspberry Pi Algorithm
   - Vercel Implementation and Supabase Integration
5. Results
6. Conclusions

Raspberry Pi Algorithm

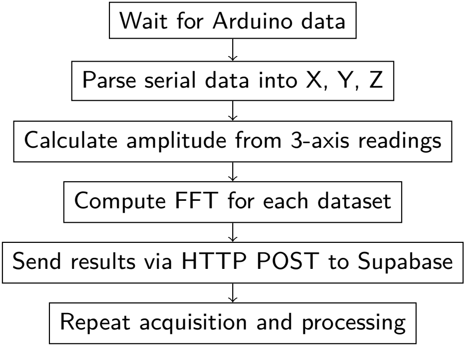


Figure 6: The Raspberry Pi 4 Algorithm: serial data reception, FFT processing, and cloud transmission.

1 Introduction

2 Background

3 Engineering and Design

4 Implementation
- Arduino UNO Algorithm
- Raspberry Pi Algorithm
- Vercel Implementation and Supabase Integration

5 Results

6 Conclusions

Vercel and Supabase Algorithmic Flow

React frontend requests vibration data via Next.js API endpoint

Next.js API receives request Uses Supabase credentials Queries database

Next.js API returns JSON response containing X, Y, Z, Magnitude measurements

React stores JSON data in local state

Extracts healthy vs current measurements Combines frequency and amplitude arrays

Generates dynamic plots using Recharts Updates plots on new data

Figure 7: Flow diagram of the Vercel and Supabase algorithm: from frontend request to dynamic visualization.

1. Introduction
2. Background
3. Engineering and Design
4. Implementation
5. **Results**
6. Conclusions

System Verification and Validation

- End-to-end checks were performed across the entire system: data acquisition from ADXL335 sensors via Arduino UNO, serial transmission to Raspberry Pi 4, processing and posting to Supabase, and frontend visualization in Vercel/React.

System Verification and Validation

- End-to-end checks were performed across the entire system: data acquisition from ADXL335 sensors via Arduino UNO, serial transmission to Raspberry Pi 4, processing and posting to Supabase, and frontend visualization in Vercel/React.
- Stress tests confirmed reliable operation under expected data loads.

System Verification and Validation

- End-to-end checks were performed across the entire system: data acquisition from ADXL335 sensors via Arduino UNO, serial transmission to Raspberry Pi 4, processing and posting to Supabase, and frontend visualization in Vercel/React.
- Stress tests confirmed reliable operation under expected data loads.
- Verified no data loss, correct timing, and accurate real-time plotting.

System Verification and Validation

- End-to-end checks were performed across the entire system: data acquisition from ADXL335 sensors via Arduino UNO, serial transmission to Raspberry Pi 4, processing and posting to Supabase, and frontend visualization in Vercel/React.
- Stress tests confirmed reliable operation under expected data loads.
- Verified no data loss, correct timing, and accurate real-time plotting.
- System performance validated for FFT processing, transmission rates, and visualization updates.

System Verification and Validation

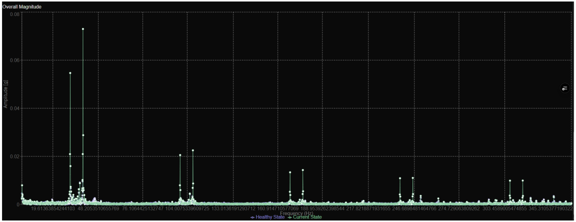


Figure 8: Visualization of the overall vibration magnitude. This chart combines all three coordinate axes into a single representation, showing the total vibration energy with the healthy baseline (blue) compared against the current state (green).

1 Introduction

2 Background

3 Engineering and Design

4 Implementation

5 Results

6 Conclusions

System Overview and Achievements

- Developed and validated an end-to-end condition monitoring system for bearing fault detection using vibration analysis.

System Overview and Achievements

- Developed and validated an end-to-end condition monitoring system for bearing fault detection using vibration analysis.
- Integrated hardware (ADXL335, Arduino UNO, Raspberry Pi 4) with cloud services (Supabase, Vercel) for data acquisition, processing, and visualization.

System Overview and Achievements

- Developed and validated an end-to-end condition monitoring system for bearing fault detection using vibration analysis.
- Integrated hardware (ADXL335, Arduino UNO, Raspberry Pi 4) with cloud services (Supabase, Vercel) for data acquisition, processing, and visualization.
- FFT-based comparative analysis of X, Y, Z axes and magnitude against healthy baselines for fault identification.

System Overview and Achievements

- Developed and validated an end-to-end condition monitoring system for bearing fault detection using vibration analysis.
- Integrated hardware (ADXL335, Arduino UNO, Raspberry Pi 4) with cloud services (Supabase, Vercel) for data acquisition, processing, and visualization.
- FFT-based comparative analysis of X, Y, Z axes and magnitude against healthy baselines for fault identification.
- Arduino collects and transmits triaxial acceleration in batches at 700 [Hz] sampling rate without loss; Raspberry Pi converts to frequency-domain and posts to Supabase.

System Overview and Achievements

- Developed and validated an end-to-end condition monitoring system for bearing fault detection using vibration analysis.
- Integrated hardware (ADXL335, Arduino UNO, Raspberry Pi 4) with cloud services (Supabase, Vercel) for data acquisition, processing, and visualization.
- FFT-based comparative analysis of X, Y, Z axes and magnitude against healthy baselines for fault identification.
- Arduino collects and transmits triaxial acceleration in batches at 700 [Hz] sampling rate without loss; Raspberry Pi converts to frequency-domain and posts to Supabase.
- Vercel dashboard visualizes frequency spectra interactively for real-time monitoring.